

The Mathematics Modeling For The Bees Population Growth To The Amount Of Honey Production (Based On Qur'an Surah An-Nahl:69)

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Abstract

The World Health Organization has reported an ongoing increase in COVID-19 cases and deaths. Surah An-Nahl verse 69 states that honey produced by bees functions as a medicinal treatment that promotes human health. Several research studies have investigated and reported that honey can improve human immune and stamina, and prevent viral infections. This study focuses on the bee population model and its relation to honey production, utilizing the Khoury model method. The objective of this study is to analyze the correlation between Surah An-Nahl verse 69 and bees, predict and optimize honey-producing bee populations, develop a bee population breeding model, and assess its benefits. The results of this research will enable the prediction of proportional bee population growth to honey production.

Keywords: 1. Bee, 2. Covid, 2. Honey, 3. Al-Qur'an, and 3. Population.

1. Introduction

COVID-19 is a highly contagious disease caused by the SARS-CoV-2 virus. The symptoms vary from mild to severe, with some people remaining asymptomatic.

The pandemic continues to affect the world, and efforts to contain it are ongoing through vaccination campaigns, social distancing, and other measures. As of May 5th, 2023, there have been over 427 million cases and 9.4 million

deaths reported worldwide, according to the World Health Organization. Researchers are working to develop effective treatments and vaccines for COVID-19, while health professionals continue to provide care and support for those affected by the disease.

Honey has been a part of human history for thousands of years, and its use in traditional medicine has been well-documented across cultures. Recent scientific studies have also shown that honey has a range of therapeutic properties, including antibacterial, anti-inflammatory, antioxidant, and wound-healing effects. Thus, honey has potential as an alternative treatment for COVID-19.

In Surah Al-Nahl, Verse 69, Allah mentions that from the belly of bees, honey of varying colors and healing for mankind is produced. This verse emphasizes the natural medicinal properties of honey that have been recognized by humans for thousands of years, as well as its variety in color and taste.

Therefore, we chose Bee Populations as the object of research because the good preservation of bee populations can result in good honey production quantities, which can be beneficial for human health.

2. Method and Experimental Details

2.1 Population

The bee population consists of various types of hive bees and

foragers bees. Bees are crucial in the ecosystem and play an important role in plant pollination, maintaining environmental sustainability, and also as a source of food such as honey and wax.

Bee populations can vary depending on environmental factors such as temperature, rainfall, climate, and habitat conditions. Bee populations can decline due to environmental changes that damage their habitat, pesticide use, and infection with diseases.

In a honey bee colony, the population consists of a queen, workers, and drones. The queen is responsible for laying eggs and ensuring the colony's survival, while the workers are female bees that collect nectar, produce honey, care for the nest and larvae, and perform pollination. Drones are usually present in small numbers and play a role in colony reproduction.

Maintaining a healthy and stable bee population is crucial to maintaining ecosystem sustainability. Some ways to maintain bee populations include providing a healthy and safe environment, reducing the use of pesticides and harmful chemicals, and practicing sustainable agriculture.

2.2 Population Model

Mathematical modeling is an approach that uses mathematical equations and

formulas to explain or predict phenomena in the real world. Mathematical modeling can be used in various fields, including modeling populations. Population modeling aims to understand and predict changes in the number and structure of populations over a certain period of time.

Khoury model method presents a mathematical model to predict the population dynamics of honey bee colonies. The model takes into account factors such as the size and age distribution of bees, brood development, foraging behavior, and environmental factors. The model was validated through comparisons with field data and simulations, and was shown to accurately predict colony growth and decline. This study contributes to our understanding of the factors that influence honey bee populations and provides a tool for beekeepers and researchers to monitor and manage honey bee colonies.

From Khoury model method we can represent a differential equation model :

$$\begin{aligned}\frac{dB_u}{dt} &= \Pi - \alpha B_u - \mu_u B_u \\ \frac{dB_c}{dt} &= \alpha B_u - \beta B_c - \mu_c B_c \\ \frac{dH}{dt} &= \beta B_c - HR(H, F) - \mu_h H \\ \frac{dF}{dt} &= HR(H, F) - \mu_f F\end{aligned}$$

There are several parameters as follows:

- B_u is uncapped brood population (eggs and larvae).
- B_c is capped brood population (pupae).
- Π is recruitment rate, the number of eggs laid daily by the queen.
- α is the rate of uncapped brood changes to pupae per day.
- β is the rate that young bees emerge from pupation per day.
- $R(H, F)$ is the recruitment rate functions, describes the effect of social inhibition on the recruitment rate.
- μ_u is the rate that uncapped broods die per day.
- μ_c is the rate that capped broods die per day.
- μ_h is the rate that hive bees die per day.
- μ_f is the rate that foragers bees die per day.

So we can simplify it to construction model in the following diagram below.

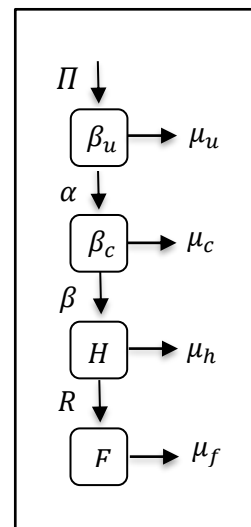


Figure 1. Compartement Diagram

The model exclusively accounts for the population of female worker bees, as male bees (drones) do not contribute significantly to the colony's dynamics, given their population generally comprises less than 5% of the total colony population. The population of bees involved in the model is divided into four stages, following the bee life cycle. It begins with the Uncapped Brood (Bu) population, consisting of eggs and larvae. The next stage is the Capped Brood (Bc) population, which comprises pupae. Finally, the adult worker bee population is divided into two distinct groups, Hive (H) and Forager (F), without any overlap between these two behavioural classes.

3. Result and Discussion

3.1 Finding Solution of Differential Equations

Differential equations (1) :

$$\begin{aligned}\frac{dB_u}{dt} &= \Pi - \alpha B_u - \mu_u B_u \\ dB_u &= (\Pi - \alpha B_u - \mu_u B_u) dt \\ \int \frac{1}{\Pi - \alpha B_u - \mu_u B_u} B_u &= \int dt \\ \ln(\Pi - \alpha B_u - \mu_u B_u) \frac{1}{-\alpha - \mu_u} &= t + c \\ \ln(\Pi - \alpha B_u - \mu_u B_u) &= (t + c)(-\alpha - \mu_u) \\ \ln(\Pi - \alpha B_u - \mu_u B_u) &= -dt - \mu_u t - \alpha c - \mu_u c \\ \Pi - \alpha B_u - \mu_u B_u &= e^{-dt} \cdot e^{-\mu_u t} \cdot e^{-\alpha c} \cdot e^{-\mu_u c} \\ \Pi - \alpha B_u - \mu_u B_u &= e^{-dt} \cdot e^{-\mu_u t} \cdot (e^c)^{-\alpha} \cdot (e^c)^{-\mu_u}\end{aligned}$$

$$\begin{aligned}\Pi - \alpha B_u - \mu_u B_u &= e^{-dt} \cdot e^{-\mu_u t} \cdot c^{-\alpha} \cdot c^{-\mu_u} \\ -\alpha B_u - \mu_u B_u &= e^{-dt} \cdot e^{-\mu_u t} \cdot c^{-\alpha} \cdot c^{-\mu_u} - \Pi \\ \alpha B_u + \mu_u B_u &= \Pi - e^{-dt} \cdot e^{-\mu_u t} \cdot c^{-\alpha} \cdot c^{-\mu_u} \\ B_u(\alpha + \mu_u) &= \Pi - e^{-dt} \cdot e^{-\mu_u t} \cdot c^{-\alpha} \cdot c^{-\mu_u} \\ B_u &= \frac{\Pi - e^{-dt} \cdot e^{-\mu_u t} \cdot c^{-\alpha} \cdot c^{-\mu_u}}{\alpha + \mu_u}\end{aligned}$$

$$\text{find } c, B_{u(0)} = B_{u0}$$

$$\begin{aligned}B_{u(0)} &= \frac{\Pi - e^0 \cdot e^0 \cdot c^{-\alpha} \cdot c^{-\mu_u}}{\alpha + \mu_u} \\ B_{u0} &= \frac{\Pi - c^{-\alpha-\mu_u}}{\alpha + \mu_u} \\ B_{u0}(\alpha + \mu_u) &= \Pi - c^{-\alpha-\mu_u} \\ c^{-\alpha-\mu_u} &= \Pi - B_{u0}(\alpha + \mu_u) \\ B_u &= \frac{\Pi - e^{-\alpha t - \mu_u t} \cdot c^{-\alpha-\mu_u}}{\alpha + \mu_u} \\ B_u &= \frac{\Pi - e^{-\alpha t - \mu_u t} \cdot (\Pi - B_{u0}(\alpha + \mu_u))}{\alpha + \mu_u}\end{aligned}$$

The formula for calculating the quantity of Uncapped Brood (B_u) is obtained, requiring B_{u0} , which represents the initial count of uncapped brood.

Differential equation (2) :

$$\frac{dB_c}{dt} = \alpha B_u - \beta B_c - \mu_c B_c$$

$$\frac{dB_c}{dt} = \alpha B_u - (\beta + \mu_c) B_c$$

$$\frac{dB_c}{dt} + (\beta + \mu_c) B_c = \alpha B_u$$

Integration factor

$$I = e^{\int (\beta + \mu_c) dt}$$

$$I = e^{(\beta + \mu_c)t}$$

$$e^{(\beta + \mu_c)t} \frac{dB_c}{dt} + e^{(\beta + \mu_c)t} (\beta + \mu_c) B_c = e^{(\beta + \mu_c)t} \alpha B_u$$

$$\frac{d}{dt} (e^{(\beta + \mu_c)t} B_c) = e^{(\beta + \mu_c)t} \alpha B_u$$

$$d(e^{(\beta + \mu_c)t} B_c) = e^{(\beta + \mu_c)t} \alpha B_u dt$$

$$\int d(e^{(\beta + \mu_c)t} B_c) = \int e^{(\beta + \mu_c)t} \alpha B_u dt$$

$$\begin{aligned}
&= \int e^{(\beta+\mu_c)t} \alpha \left[\frac{e^{(\beta+\mu_c)t} B_c}{\alpha + \mu_u} (\Pi - e^{-(\alpha-\mu_u)t} (\Pi - B_{u0}(\alpha + \mu_u))) \right] dt \\
&e^{(\beta+\mu_c)t} B_c = \frac{\alpha}{\alpha + \mu_u} \int e^{(\beta+\mu_c)t} (\Pi - e^{-(\alpha-\mu_u)t} (\Pi - B_{u0}(\alpha + \mu_u))) dt \\
&e^{(\beta+\mu_c)t} B_c = \frac{\alpha}{\alpha + \mu_u} \int [e^{(\beta+\mu_c)t} \Pi - e^{(\beta+\mu_c-\alpha+\mu_u)t} (\Pi - B_{u0}(\alpha + \mu_u))] dt \\
&= \frac{\alpha}{\alpha + \mu_u} \left[\frac{e^{(\beta+\mu_c)t} \Pi}{\beta + \mu_c} - \frac{(\Pi - B_{u0}(\alpha + \mu_u)) e^{(\beta+\mu_c-\alpha+\mu_u)t}}{\beta + \mu_c - \alpha + \mu_u} + C \right] \\
&B_c = \frac{\alpha}{\alpha + \mu_u} \left[\frac{\Pi}{\beta + \mu_c} - \frac{(\Pi - B_{u0}(\alpha + \mu_u)) e^{-(\alpha-\mu_u)t}}{\beta + \mu_c - \alpha + \mu_u} + C \right]
\end{aligned}$$

The formula for calculating the quantity of Capped Brood (B_c) is obtained, where C represents a constant dependent on the initial value of B_c .

Differential equation (3) :

$$\frac{dH}{dt} = \beta B_c - HR(H, F) - \mu_h H$$

$$\frac{dH}{dt} = \beta B_c - HR - \mu_h H$$

$$\frac{dH}{dt} = \beta B_c - HR - \mu_h H$$

$$\frac{dH}{dt} = \beta B_c - (R + \mu_h) H$$

$$\frac{dH}{dt} + (R + \mu_h) H = \beta B_c$$

Integration factor

$$I = e^{\int (R+\mu_h) dt}$$

$$I = e^{(R+\mu_h)t}$$

$$e^{(R+\mu_h)t} \frac{dH}{dt} + e^{(R+\mu_h)t} (R + \mu_h) H = e^{(R+\mu_h)t} \beta B_c$$

$$\frac{d}{dt} (e^{(R+\mu_h)t} H) = e^{(R+\mu_h)t} \beta B_c$$

$$d(e^{(R+\mu_h)t} H) = e^{(R+\mu_h)t} \beta B_c dt$$

$$\int d(e^{(R+\mu_h)t} H) = \int e^{(R+\mu_h)t} \beta B_c dt$$

$$\begin{aligned}
e^{(R+\mu_h)t} H &= \int e^{(R+\mu_h)t} \beta \frac{\alpha}{\alpha + \mu_u} \left[\frac{\Pi}{\beta + \mu_c} - \frac{(\Pi - B_{u0}(\alpha + \mu_u)) e^{-(\alpha-\mu_u)t}}{\beta + \mu_c - \alpha + \mu_u} + C \right] dt
\end{aligned}$$

$$\begin{aligned}
e^{(R+\mu_h)t} H &= \frac{\alpha \beta}{\alpha + \mu_u} \int e^{(R+\mu_h)t} \left[\frac{\Pi}{\beta + \mu_c} - \frac{(\Pi - B_{u0}(\alpha + \mu_u)) e^{-(\alpha-\mu_u)t}}{\beta + \mu_c - \alpha + \mu_u} + C \right] dt
\end{aligned}$$

$$\begin{aligned}
&= \frac{\alpha \beta}{\alpha + \mu_u} \int \left[e^{(R+\mu_h)t} \frac{\Pi}{\beta + \mu_c} - \frac{(\Pi - B_{u0}(\alpha + \mu_u)) e^{(R+\mu_h-\alpha+\mu_u)t}}{\beta + \mu_c - \alpha + \mu_u} + C \right] dt
\end{aligned}$$

$$\begin{aligned}
&= \frac{\alpha \beta}{\alpha + \mu_u} \left[e^{(R+\mu_h)t} \frac{\Pi}{(R + \mu_h)(\beta + \mu_c)} - \frac{(\Pi - B_{u0}(\alpha + \mu_u)) e^{(R+\mu_h-\alpha+\mu_u)t}}{(R + \mu_h - \alpha + \mu_u)(\beta + \mu_c - \alpha + \mu_u)} + C_1 + C_2 \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{\alpha \beta}{\alpha + \mu_u} \left[\frac{\Pi}{(R + \mu_h)(\beta + \mu_c)} - \frac{(\Pi - B_{u0}(\alpha + \mu_u)) e^{-(\alpha-\mu_u)t}}{(R + \mu_h - \alpha + \mu_u)(\beta + \mu_c - \alpha + \mu_u)} + C_1 + C_2 \right]
\end{aligned}$$

The formula for calculating the quantity of Hive Bees (H) is obtained, where C_1 represents a constant dependent on the initial value of B_c and C_2 represents a constant dependent on the initial value of H .

Differential equation (4) :

$$\frac{dF}{dt} = HR(H, F) - \mu_f F$$

$$\frac{dF}{dt} = HR - \mu_f F$$

$$\frac{dF}{dt} + \mu_f F = HR$$

Integration factor

$$I = e^{\int \mu_f dt}$$

$$I = e^{\mu_f t}$$

$$e^{\mu_f t} \frac{dF}{dt} + e^{\mu_f t} \mu_f F = e^{\mu_f t} HR$$

$$\frac{d}{dt} (e^{\mu_f t} F) = e^{\mu_f t} HR$$

$$\begin{aligned}
d(e^{\mu_f t})F &= (e^{\mu_f t}HR)dt \\
\int d(e^{\mu_f t})F &= \int (e^{\mu_f t}HR)dt \\
&= \frac{R\alpha\beta}{\alpha + \mu_u} \int e^{\mu_f t} \left[\frac{\Pi}{(R + \mu_h)(\beta + \mu_c)} \right. \\
&\quad \left. - \frac{(\Pi - B_{u0}(\alpha + \mu_u))e^{-(\alpha - \mu_u)t}}{(R + \mu_h - \alpha + \mu_u)(\beta + \mu_c - \alpha + \mu_u)} + C_1 + C_2 \right] dt \\
&= \frac{R\alpha\beta}{\alpha + \mu_u} \int \left[\frac{e^{\mu_f t} \Pi}{(R + \mu_h)(\beta + \mu_c)} \right. \\
&\quad \left. - \frac{(\Pi - B_{u0}(\alpha + \mu_u))e^{\mu_f - (\alpha - \mu_u)t}}{(R + \mu_h - \alpha + \mu_u)(\beta + \mu_c - \alpha + \mu_u)} + C_1 + C_2 \right] dt \\
&= \frac{R\alpha\beta}{\alpha + \mu_u} \left[\frac{e^{\mu_f t} \Pi}{\mu_f (R + \mu_h)(\beta + \mu_c)} \right. \\
&\quad \left. - \frac{(\Pi - B_{u0}(\alpha + \mu_u))e^{\mu_f - (\alpha - \mu_u)t}}{(\mu_f - \alpha + \mu_u)(R + \mu_h - \alpha + \mu_u)(\beta + \mu_c - \alpha + \mu_u)} \right. \\
&\quad \left. + C_1 + C_2 + C_3 \right] \\
&= \frac{R\alpha\beta}{\alpha + \mu_u} \left[\frac{\Pi}{\mu_f (R + \mu_h)(\beta + \mu_c)} \right. \\
&\quad \left. - \frac{(\Pi - B_{u0}(\alpha + \mu_u))e^{-(\alpha - \mu_u)t}}{(\mu_f - \alpha + \mu_u)(R + \mu_h - \alpha + \mu_u)(\beta + \mu_c - \alpha + \mu_u)} \right. \\
&\quad \left. + C_1 + C_2 + C_3 \right]
\end{aligned}$$

The formula for calculating the quantity of Foragers Bees (F) is obtained, where C_3 represents a constant dependent on the initial value of F .

3.2 Honey Production

Honey beekeeping is an activity of honey bee management or maintenance which aims to obtain the expected bee products/products. While product harvesting is the activity of collecting or harvesting products produced from honey bees. The products produced include: honey, propolis, bee pollen, royal jelly, wax/wax, bee stings. Honey is a viscous

liquid produced by honey bees from various nectar sources that still have diastase enzyme activity. The amount of honey production is influenced by the number of bee populations because usually the number of populations is proportional to the amount of honey production (proportional). The parable is:

$$M = aF$$

Description:

M = Number of honey production

a = Conversion parameter with assumption that 10.000 bees can produce 1 Kg honey

F = Number of bees

3.3 Numerical Simulation

Numerical simulation was performed with initial values $t_0 = 0$, $t_f = 90$, $dt = 1$, and initial conditions (eggs, larva, pupae, adult worker bees) $x_0 = (30,0,10,10)$. In addition, the values of the parameters used are $\Pi = 0,1$, $\alpha = 0,05$, $\beta = 0,1$, $\mu_f = 0,2$.

Next, a graph of Bees Populations is presented in the following figure.

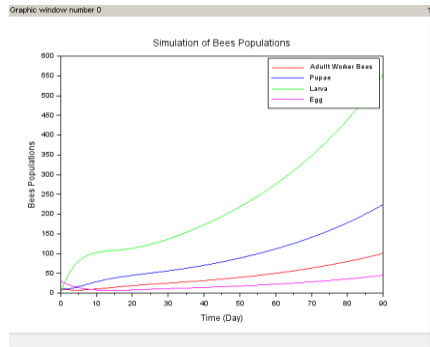


Figure 2. Graph of Bees Populations.

And graph of Honey Production is presented in the following figure.

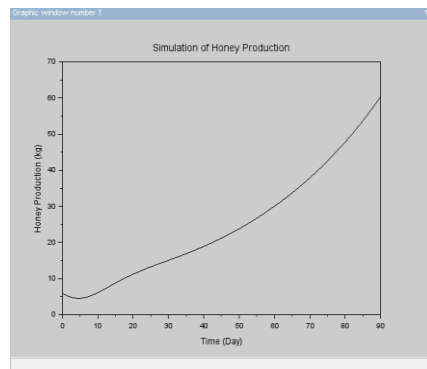


Figure 2. Graph of Honey Productions.

This program is a simple implementation of the Khoury model method in modeling the life cycle of bees from eggs to adult bees and producing honey. This model has two age classes, namely egg and adult bees, with a different reproduction number r for the age of each class. The program evaluates bee populations over time, which consists of the reproduction number and the probability of living out the life of each class. Then, using the same function, the amount of honey produced each time is also calculated. The output of the

program is in the form of a graph showing the bee population and the amount of honey produced at any time throughout the bee life cycle. In the graph it can be seen that the amount of honey produced increases with the increase in the number of adult bees.

3.4 Analysis of the relationship of the Qur'anic verses

(QS. An Nahl : 69)

ثُمَّ كُلِّي مِنْ كُلِّ النَّمْرَاتِ فَاسْلُكِي سُبُلَ رَبِّكِ ذُلُلًا
يَخْرُجُ مِنْ بُطُونِهَا شَرَابٌ مُخْتَلِفٌ أَلْوَانُهُ فِيهِ شِفَاءٌ
لِلنَّاسِ إِنَّ فِي ذَلِكَ لَآيَةً لِقَوْمٍ يَتَفَكَّرُونَ

Explain:

“and feed from ‘the flower of’ any fruit ‘you please’ and follow the ways your Lord has made easy for you.” From their bellies comes forth liquid of varying colours, in which there is healing for people. Surely in this is a sign for those who reflect.”

The verse implies that bees, as one of Allah's creations, have great benefits for humans. Honey, which is produced by bees, has medicinal properties that are healing for humans. This indicates the greatness of Allah, who has created bees creatures that are useful to humans. The verse also teaches humans to pay attention to the signs of Allah's greatness that are found in His creation.

It is not explained in detail what diseases honey can cure. However, honey is generally beneficial for producing energy, increasing immunity, and improving stamina. Furthermore, honey contains acetylcholine,

which can improve metabolism by improving blood circulation and lowering blood pressure.

Despite its low pH, honey can actually increase the pH of the stomach. This is due to the alkaline minerals found in honey that function as a buffer. Moreover, the Fe content in honey can increase the number of red blood cells in human blood and increase hemoglobin levels. Honey is also often used to treat canker sores, uric acid, heart disease, asthma, and high cholesterol levels.

Honey contains antibacterial substances and has hygroscopic properties, which means it can attract water from the surrounding environment. Therefore, it is good for treating external wounds and infectious diseases. For body care, honey can be made into lotion, masks, soaps, shampoos, and body scrubs. For skin care, honey can be used as a powder. For eye health, honey is used as eye drops. In the cosmetics industry, honey is used as a component of lipstick, moisturizers, and skin antiseptics.

Based on the explanation in the Qur'an in Surah An-Nahl: 69 and several studies on the benefits of honey, it is necessary for us to preserve and research the stability and development of bee populations. The good preservation of bee populations can result in good honey production quantities, which can be beneficial for human health.

4. Conclusion

The bee life cycle starts from a larva to pupae, then develops into young bees or forager bees, when they are old enough the bees will look for honey and pollen and then live up to 5 or 6 weeks

From the model, it can be concluded that over time the bee population will continue to increase, and of course honey production will also increase in proportion to the number of bees.

Thus with the large amount of honey production, the honey can be consumed by the community and increase the immune system.

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